LLMS IN SOFTWARE ENGINEERING

# RocketRewards: The AI Financial & Card Copilot

Re-engineering Rocket Money into an autonomous credit rewards optimizer and debit-style debt guardrail.

**Deterministic LLM Tool Calling** **Safe-Spend Guardrails** **Zero-Breakage Routing**

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RUBRIC 1 · PROBLEM & MARKET OPPORTUNITY

# The Pain Point & Target User

**$3.4B**

### Total Addressable Waste

Unredeemed card perks and mismatched checkout swipes in the US annually.

**$420**

### Lost Value Per Cardholder

Average annual breakage from expired statement credits and suboptimal category choices.

**68%**

### Beginner Debt Anxiety

Gen-Z and college graduates report fear of APR penalties and accidental score drops.

### Target Segment A: Reward Optimizers

Young professionals holding 3–5 premium cards suffering from merchant category confusion.

### Target Segment B: First-Time Cardholders

Students and young earners needing debit-style balance protection to establish prime scores.

RUBRIC 1 · COMPETITIVE DIFFERENTIATION

# Bridging the Gap: Market Opportunity & Current Solutions

| Platform | Budgeting & Subs | Swipe Advisor | Debt Guardrails | Core Structural Weakness |
|---|---|---|---|---|
| **Rocket Money** | **Comprehensive** | None | None | Reactive backward-looking expense categorization |
| **Credit Karma** | Minimal | Sponsored Links | Basic Score View | Monetizes high-interest credit card lead volume |
| **CardPointers** | None | **Static Multipliers** | None | Blind to checking account liquidity and credit utilization |
| **RocketRewards** | **AI Subscriptions** | **Contextual Copilot** | **Deterministic Caps** | **Full-stack proactive financial agent** |

**Strategic Defensibility:** Combines live balance awareness with credit card benefit logic to solve point-of-sale decisions.

RUBRIC 2 · CORE VALUE PROPOSITION

# What We Do: Proactive Guidance Powered by Spending History

**The Direct Solution:**
RocketRewards combines historical spending analytics with real-time checkout intelligence to eliminate reward breakage and safeguard credit health.

### History-Driven Card Match

Analyzes your past 6–12 months of transactions to recommend the exact cards that maximize net returns on your specific lifestyle spending.

**Solves: Misaligned cards & spammy affiliate offers**

### Active Reward & Perk Routing

Pairs real-time POS multiplier routing with automated tracking of monthly dining, travel, and retail credits before they expire.

**Solves: $420+ annual reward & credit forfeiture**

### Safe-Spend Guardrails

Checks live checking reserves and locks utilization below 10%, scheduling mid-cycle payments to keep beginner scores growing.

**Solves: First-timer debt & credit score drops**

**From retroactive categorization to proactive decision-making at checkout.**

Zero Hallucination · Deterministic Execution

RUBRIC 2 · SCOPE BOUNDARIES & HYPOTHESIS

# Testable Core Hypothesis & Bounded MVP Scope

TESTABLE CORE HYPOTHESIS

**"If users receive automated point-of-sale card recommendations with safe-spend caps, at least 65% will adopt the recommended card, reducing annual reward breakage by over $300."**

### In-Scope MVP Modules

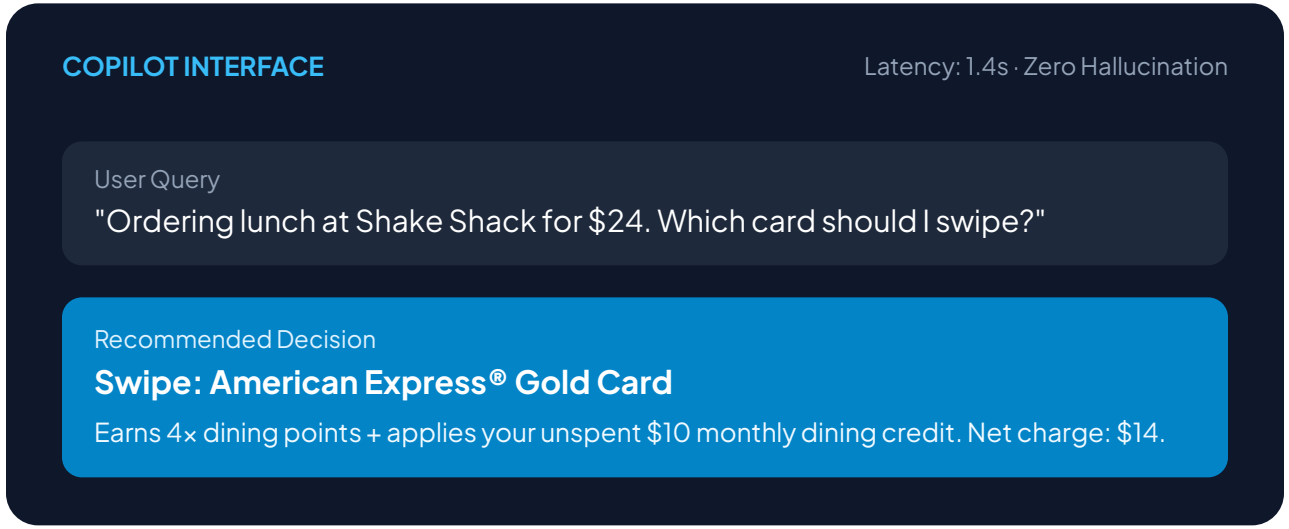
- Natural Language "Which Card?" point-of-sale routing
- Deterministic Safe-Spend calculator protecting checking buffer
- Expiring statement credit tracker across enrolled wallets

### Explicit Out-of-Scope Boundaries

- Production Plaid API integrations (simulated via synthetic ledgers)
- Automated bill cancellation calling agents
- Live credit bureau hard-inquiries or card underwriting

RUBRIC 2 · VISUAL PRODUCT DEMO

# Scenario A: Point-of-Sale Rewards Maximization




### 1. Merchant Category Code (MCC) Resolution

Maps Shake Shack to Fast Casual Dining (MCC 5814) rather than generic retail.

### 2. Benefit Expiration Tracking

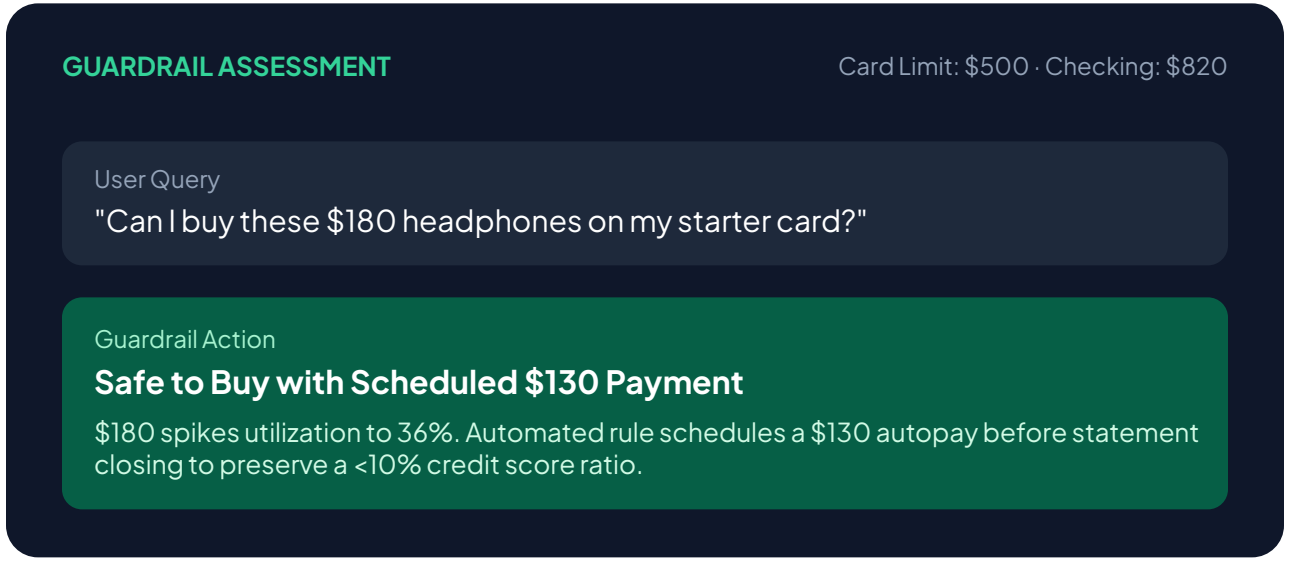
Flags $10 monthly credit expiring in 3 days, saving $120 per year in unredeemed perks.

### 3. Automated Wallet Ranking

Outperforms flat 2% card alternative by delivering an effective 43% savings on this order.

RUBRIC 2 · VISUAL PRODUCT DEMO

# Scenario B: Safe-Spend Guardrails for First-Timers




### 1. Debit Parity Verification

Confirms liquid checking reserve ($820) exceeds charge without dipping below emergency buffer.

### 2. Statement vs Due Date Shield

Calculates interim payment date to prevent high balances from reporting to credit bureaus.

### 3. 750+ Credit Score Pathway

Keeps thin-file builders at 0% APR interest exposure while building credit history.

RUBRIC 3 · VALIDATION PLAN & DECISION GATES

# Actionable User Testing Plan & Quantitative Gates

CONTROLLED COHORT PROTOCOL

**N = 20 Target Users (10 Multi-Card Optimizers · 10 First-Time Cardholders)**

**Method:** 14-day simulated point-of-sale test suite (~25 transaction prompts per user)

METRIC 1

## Recommendation Adoption

% of transactions using copilot-selected card.

✓ PASS: >65% adoption
✗ PIVOT: <45% (Shift to push reminders)

METRIC 2

## Utilization Protection

% of users avoiding statement balance spikes >10%.

✓ PASS: >90% sub-10% compliance
✗ PIVOT: <75% (Enforce strict checkout block)

METRIC 3

## Engineering Accuracy

Deterministic tool precision across edge cases.

✓ PASS: 100% arithmetic accuracy
✗ PIVOT: >0% hallucinations (Refactor tool schema)

# Empowering Every Swipe with Intelligence

Maximizing returns for experienced cardholders while safeguarding first-time builders from debt.

Open for Questions & Discussion

APPENDIX & SCHOLARLY FOUNDATIONS

# References, Regulatory Data & Architecture Attribution

### ⚖ Reward Breakage & Waste

**CFPB (May 2024 / Circular 2024-07):** *Credit Card Rewards Report & Issue Spotlight*. Documents 90%+ spending on rewards cards, $40B+ earned annually, and ~$500M directly forfeited by consumers each year in breakage and opaque expirations.

**Bond Brand Loyalty (2024):** Quantified 44% of loyalty cardholders forfeit points due to expiration windows; estimated $100B+ global unredeemed liability balances under ASC 606.

Validates Slide 2 & Slide 3 Pain Points

### Debt & Utilization Traps

**Federal Reserve Bank of New York (2024):** *Quarterly Report on Household Debt & Credit*. Documents aggregate US credit card balances surpassing $1.14T, with delinquency transitions disproportionately hitting cardholders aged 18–29 with utilization >30%.

**CFPB CARD Act Biennial Report (2023–2025):** Found average APR margins of 15.4% above prime (~22–25% total), erasing 100% of reward value whenever revolving debt is carried across statement cycles.

Validates Slide 5 Hypothesis & Slide 7 Guardrails

### </> Software Architecture & RAG

**Lewis, P., et al. (NeurIPS 2020):** *"Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks."* Architectural basis for grounding unstructured merchant queries into vector embeddings for MCC category retrieval.

**LangChain / LangGraph & Pydantic V2:** Stateful cyclic agent graphs combined with strict schema enforcement to guarantee deterministic execution of balance arithmetic without generative hallucinations.

Validates Slide 4 Engineering & Slide 8 Gate

**All financial figures, APR margins, and delinquency statistics are derived from official federal registries & peer-reviewed CS literature.**